

Mini Capstone Project: Airline Operations Analytics Platform using Azure Databricks

Business Scenario

SkyJet Airways wants to build a modern analytics platform on Azure Databricks to analyze:

- Flights
- Passengers
- Bookings
- Revenue
- Delays
- Routes
- Daily incremental loads

The objective is to build an end-to-end Delta Lake solution.

Dataset 1: flights.csv

```
flight_id
airline
from_city
to_city
duration
status
```

Example:

```
F101,Indigo,Hyderabad,Delhi,140,On Time
F102,Air India,Mumbai,Chennai,120,Delayed
```

Dataset 2: bookings.csv

```
booking_id
flight_id
passenger_name
travel_class
ticket_price
booking_date
```

Example:

```
B1001,F101,Rahul Sharma,Economy,8500,2026-06-01
B1002,F101,Priya Reddy,Business,22000,2026-06-01
```

Dataset 3: passenger_preferences.json

```
[
{
  "passenger_name":"Rahul Sharma",
  "meal":"Veg",
  "seat":"Window"
},
{
  "passenger_name":"Priya Reddy",
  "meal":"Non-Veg",
  "seat":"Aisle"
}
]
```

Part 1 : Data Ingestion

Create notebooks:

notebook_1_ingest_flights

```
Read flights.csv
Infer schema
```

Display schema

notebook_2_ingest_bookings

Read bookings.csv

Display data

notebook_3_ingest_preferences

Read JSON

Flatten JSON

Part 2 : Exploratory Data Analysis

Generate:

Revenue by Airline

Bar Chart

Revenue by Travel Class

Pie Chart

Flights by Status

Pie Chart

Top Routes

Horizontal Bar Chart

Ticket Price Distribution

Scatter Plot

Use:

```
matplotlib
```

Part 3 : Transformations

Create:

```
revenue = ticket_price
```

```
price_band
```

```
>20000 Premium
```

```
>10000 Standard
```

```
Else Budget
```

Create:

```
delay_flag
```

```
Delayed -> Yes
```

```
Else No
```

Part 4 : Joins

Join:

```
Flights
```

```
Bookings
```

```
Preferences
```

Generate:

```
Complete Passenger Journey Dataset
```

Part 5 : Spark SQL

Create:

```
createOrReplaceTempView()
```

Write SQL queries:

```
Revenue by airline
```

```
Revenue by route
```

```
Average ticket price
```

```
Most popular destination
```

Part 6 : Window Functions

Find:

```
Top 3 revenue flights
```

```
Top routes by airline
```

```
Running revenue
```

```
Rank airlines
```

```
Dense Rank destinations
```

Part 7 : Delta Lake

Method 1

Create Delta table using:

```
.save()
```

Method 2

Create Delta table using:

```
saveAsTable()
```

Method 3

Create Delta table using SQL

```
CREATE TABLE ...  
USING DELTA
```

Part 8 : Incremental Load

Day 1:

```
100 bookings
```

Day 2:

```
10 new bookings  
5 price updates
```

Implement:

```
MERGE INTO
```

Perform:

Part 9 : Time Travel

Read:

```
Version 0
```

```
Version 1
```

```
Latest Version
```

Compare:

```
Before merge
```

```
After merge
```

Part 10 : Optimize Delta

Run:

```
OPTIMIZE booking_master
```

Run:

```
OPTIMIZE booking_master  
ZORDER BY(flight_id)
```

Run:

```
VACUUM booking_master
```

View:

Part 11 : Tables and Views

Create:

Managed Table

```
saveAsTable()
```

External Table

```
LOCATION ...
```

Temporary View

```
createOrReplaceTempView()
```

Global Temporary View

```
createOrReplaceGlobalTempView()
```

Part 12 : Job Workflow

Notebook A

```
ingest_flights
```

↓

Notebook B


```
ingest_bookings
```



Notebook C

```
transform_data
```



Notebook D

```
generate_reports
```

Configure:

- Task dependencies
- Retry count
- Timeout
- Parameters

Part 13 : Schedule

Create Job:

```
Airline Analytics Pipeline
```

Schedule:

```
Daily at 8 AM
```

Enable:

```
Retries = 2
```

Monitor:

Execution History

Logs

Task Duration

Deliverables

Bronze Layer

Raw CSV and JSON

Silver Layer

Cleaned Delta Tables

Gold Layer

Business Reports

Final Reports

Airline Revenue Report

| Airline | Revenue |

Route Performance Report

| Route | Revenue |

Passenger Preference Report

| Passenger | Meal | Seat |

Flight Delay Report

| Flight | Status |

Top Revenue Flights

| Flight | Rank |
